Lab 5 — Static Analysis Issues Table

Baseline findings and corrective actions for the scholarship screening program.

Issue / Tool	Finding	Corrective action
Bandit B307	Unsafe eval() call used to calculate the merit score.	Replace eval() with a restricted arithmetic-expression evaluator.
Bandit B110	Exception handler silently ignores errors with pass.	Catch only expected exceptions and handle them explicitly.
Pylint W0102	Mutable default argument logs=[].	Use logs=None and create a fresh list inside the function.
Pylint W1514 / R1732	Files are opened without explicit encoding/context managers.	Use with statements and UTF-8 encoding.
Flake8 F401 / Pylint W0611	Unused math and sys imports.	Remove unused imports.
Pylint C0114 / C0116	Missing module and function docstrings.	Add concise module, class, and function docstrings.
Pylint C0103 / W0622	Non-standard function naming and built-in name shadowing.	Use snake_case and rename the credits parameter.
Flake8 E302 / E305	Incorrect blank-line spacing around functions.	Follow standard Python spacing conventions.
Flake8 E712 / Pylint C0121	Comparison against True is redundant.	Test the Boolean value directly.
Input validation	Invalid or missing GPA and negative numeric inputs are not validated.	Validate GPA, credits, and income before calculating eligibility.
Pylint W0603	Global state is modified inside functions.	Keep state in a ScholarshipScreener instance.